

LONGKHAR
NWFPs
MANAGEMENT
AND
MARKETING
PLAN
(2023-2027)



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PART 1: Summary of Completion Report of Longkhar NWFP Management Plan (2018-2022)

1.1. Physical achievements

- The sale of Non-Wood Forest Products (as raw materials or finished products) helped in improving the livelihood and boosting the economy of the Longkhar Community.
- The formation of this group helped in bringing strong collaboration among all the households of the Longkhar community.
- By understanding the importance of forest/nature, the group had carried out various protection and conservation works in collaboration with Bumdeling Wildlife Sanctuary and Gewog Office.
- Upon knowing all the harvesting procedures and norms, the group harvested NWFPs following a scientific and sustainable approach. This had not just helped them earn a living for themselves, but also kept provisions for the younger generations to benefit in times to come.

1.2. Financial achievements

Table 1. Financial achievement of the group from the year 2018-2022

Period	Products	Unit	Quantity harvested	Month of harvest	Quantity sold	Price/unit (Nu.)	Total Income
2018 to 2022	<i>Daphne</i>	Kg	2161	May-June	2161	50	108050
	<i>Bamboo</i>	Pieces	70800	Aug-Mar	70800	0.5	35000
	<i>Tshar</i>	Pieces	43200	March	43200	1	43200
	<i>Wood Burr</i>	Nos	300	February	300	500	150000
	<i>Thoksampa</i>	Kg	Nil				
Total							336250

1.3. Recommendations

- The formation of the Group had not just benefited the community at large but also been able to protect forest and NWFPs resources, hence the continuity of this group is highly recommended.
- Need additional training of book keeping and group management (administrative works) for committee members.
- Need to explore more markets and plan effective marketing strategies for the group.
- The group members need training for nursery development to carry out community afforestation and reforestation programs.

PART 2: Longkhar NWFP Management and Marketing Plan (2023-2027)

2.1. Background

The village of Longkhar, located next to Shingphel, is one of the furthest and the remotest villages of Bumdeling Gewog consisting of 12 households with the total population of 114 people. Of the total population, only 30% are living outside of the village (which includes those who are monks, school/college students, civil servants, etc.), whereas 70% of the population are still residing inside. Longkhar is 5 to 6 hours walk from the Gewog Centre and 3 hours walk from the nearest farm road. The village is comparatively less developed than the lower parts of the Gewog, owing to its remote and difficult location with less arable land.

With limited number of livestock and less arable land, the products from Agriculture and livestock farming were barely sufficient for self-consumption, hence, from time immemorial, as alternative means of income, people of Longkhar have been depending mainly on Non-Wood Forest Products such as Bamboo, Daphne, Cordyceps, Thoksumpa and Wood burr for their livelihood. The people of this community have indigenous knowledge and exceptional skills for producing various bamboo products, which are battered with rice and other commodities with lower villages of the Gewog and even sell their products during local and Dzongkhag festivals. Daphne barks (raw materials for Desho paper production) and Thrashing blocks (raw materials for wood turning) are being sold to Local factories of Bumdeling and Yangtse, cordyceps and the rhizomes of Thoksumpa (high revenue fetching non wood products) are being traded with international markets. Therefore, realizing NWFPs's value and their multiple uses, finding the needs to sustainably utilize these resources, and preserving age old tradition and skills of Longkhar, the BWS and Longkhar Community collaboratively initiated and formed NWFPs Management and Marketing group in the year 2017.

Prior to the formation of NWFP management and marketing groups, until 2017, people of Longkhar had limited or no legal rights to access NWFPs, less markets for selling their products and less income generation with annual household income ranging from Nu.30,000 to Nu.60,000. Worst still, they followed traditional method of harvesting technique which was not scientific and sustainable. Whereas, ever since the establishment of NWFP Management and Marketing group, not only the link between people and forest strengthened (with more conservation efforts and sustainable utilization of resources) but also contributed hugely to the economic development of the community (by exercising the group user right to harvest and sale of NWFPs). Every household in the community now earns more than Nu.100,000 annually and the group, particularly, has earned around 500,000 within the five years period.

With tremendous benefits ripped from the previous plan period, the continuity and revision of the plan proved to be indispensable. Hence, the plan revision was carried out jointly by the Group and the BWS incorporating better scopes and strategies, where both management of resources and marketing of their products are being prioritized and planned for effective implementation.

2.2. Objectives:

This management and marketing plan aims to achieve the following objectives:

- To ensure equal distribution of natural resources among local communities through group management
- To expand the benefits of NWFPs for economic development of Longkhar community which will contribute to the poverty alleviation.
- To inculcate sense of ownership, encourage community participation and empower local community for conservation of nature and natural resources.
- To promote diverse products and organize systematic marketing of products.

2.3. Boundary Map

The Map shows the boundaries of harvesting areas of NWFPs (Daphne, Bamboo and Thoksumpa) of Longkhar, which are demarcated(mapped) based on traditional boundaries, customary practices, and availability of natural resources.

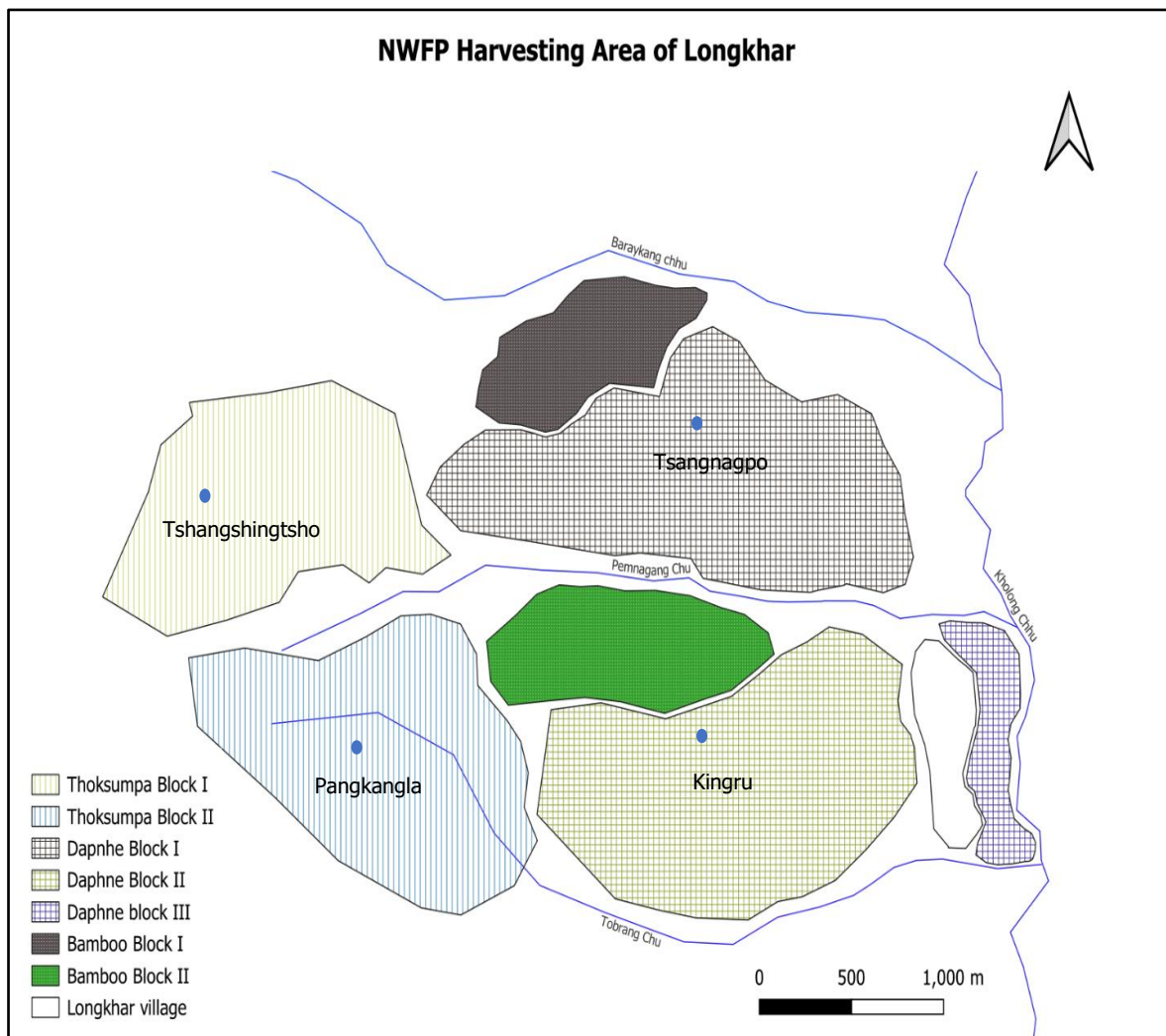


Figure 1: NWFPs harvesting area of Longkhar

2.4. Description of Blocks

2.4.1. Bamboo Blocks

Two bamboo blocks have been identified and demarcated as bamboo harvesting area and it was done based on the collection of information regarding traditional right and practices from Longkhar people and subsequent resource assessment. The first block is located at Tsangnagpo, with an area of 54.4 hectares and it takes 1 hour 30 minutes by walk to reach this block from Longkhar village. The second block is situated at Kingru with total area of 72.8 hectares and it takes just an hour to reach at this site from Longkhar village. The dominant species found in this block are *Borinda grossa*, with *Sorbus cuspidata*, *Magnolia campbellii*, *Quercus semicarpifolia*, and *Rhododendron arboretum*.

2.4.2. Daphne Blocks

Three Daphne blocks have been identified and demarcated as Daphne harvesting area and it was done based on the collection of information regarding traditional right and practices from Longkhar people and subsequent resource assessment. The first block is located at Tsangnagpo, with an area coverage of 202 hectares and it takes 1 hour, by walk, to reach this block from Longkhar village. The second block is situated at Kingru with total area of 197 hectares and only half an hour walking distance from the village. The third block is located just below Longkhar village adjacent to Kholong chhu with total area coverage of 28.3 hectares and 15 minutes of walking distance from the village. The dominant species found in this block are *Daphne bholua*, *Alnus nepalensis*, *Quercus semicarpifolia*, *Acer campbellii*, *Rhododendron arboretum* and *Symplocas* sp.

2.4.3. Thoksumpa Blocks

Two Thoksumpa blocks have been identified and demarcated as Thoksumpa harvesting area and it was done based on the collection of information regarding traditional right and practices from Longkhar people and subsequent resource assessment. The first block is located at Tshashingtsho, with an area of 146 hectares and it takes 2 hours 45 minutes by walk to reach this block from Longkhar village. The second block is situated at Pangkangla with total area of 175 hectares and 2 hours 30 minutes walking distance from Longkhar village. The dominant species found in this block are *Tsuga dumosa*, *Abies densa*, and *Rhododendron* sp.

2.4.4. Thrashing Block

Since thrashing resources (Thrashing bearing trees, *Persia* species, acer species and *Betula* species) are scattered all over Longkhar forests, therefore, whole Longkhar forest has been identified as Thrashing block. While assessing, it was found that there are ample resources, which will be sufficient for the next 10 to 15 years, provided the resources are harvested following the prescribed harvesting methods. The dominant species found in this block are *Daphne bholua*, *Alnus nepalensis*, *Quercus semicarpifolia*, *Acer campbellii*, *Rhododendron arboretum*, *Persia* sp. and *Symplocas* sp.

2.5. Description of NWFPs and Resource Assessment

2.5.1. Bamboo

There are two types of bamboos found in Longkhar namely *Borinda grossa* and *Fargesia* spp. *Borinda grossa* has long and reasonably straight culms useful for weaving roof mats, mats, baskets, hats, and other

containers, whereas *Fragesia* species are not used apart from making traditional arrows. The culms *B.grossa* can attain a height of 10m and a diameter of 8-9 cm with the length of internode reaching 50 cm. Matured culms (i.e. more than 3 years) are used for weaving bamboo mat and roofing and fencing purposes and young culms (i.e. 1 year) are used for making bamboo products such as baskets, containers, carpet etc. The harvesting of bamboo is done from March to August every year. It grows well in Longkhar having fertile soil and favorable climatic conditions.

Since time immemorial, the people of Longkhar have been depending on bamboo for their livelihood. The people of Longkhar have indigenous skills of bamboo crafts, passed down from their forefathers and so far, they could make around 20 diverse products (Winnower (Dala), Bamboo bowl (Blow), Mat carpet (Pari), Basket (Khompo), etc.) for self-usage and for bartering with other necessary commodities (such as cereals). With increasing population in Longkhar, the pressure on *Borinda grossa* kept increasing, however, the group strictly followed harvesting prescriptions and maintained good sustainability of bamboo resources.

2.5.2. Assessment of Resources

The rigorous assessment of *Borinda grossa* was carried out by adopting the methodology of DoFPs' Guidelines for Resource assessment and management of *Borinda grossa*. This was done to review and validate the boundary prepared during participatory mapping, assess the availability of resources, and estimate the resources that can be harvested on sustainable basis from the demarcated bamboo blocks. While assessing, it was found that the resources are abundant in these blocks as depicted briefly depicted in the table 2.

Table 2. Total culms and Annual Harvesting limit of bamboo in blocks

Age Class	Block I (54.4 hectares)				Block II (72.8 hectares)			
	S-block I (19 ha)	S-block II (18.4 ha)	S-block III (17 ha)	AHL	S-block I (24 ha)	S-block II (23.8 ha)	S-block III (25 ha)	AHL
1+ year old culms	24320	23552	21760	22978	30720	30464	32000	30751
2+ year old culms	32680	31648	29240	30877	41280	40936	43000	41321
3+ year old culms	41800	40480	37400	83776	52800	52360	55000	112112
Total culms in S-blocks	98800	95680	88400		124800	123760	130000	
Total AHL (33% of total stock of (1+ year + 2 + year) culms + 70% of 3+ year culms)				137632	Total AHL (33% of total stock of (1+ year + 2 + year) culms + 70% of 3+ year culms)			184184

2.5.3. Daphne

Daphne bholua is found abundantly in longkhar forest. It is a shrub of 1-4 meters in height and occurs in evergreen oak, spruce, hemlock, fir, and blue pine forests at an altitude of 1950-3400 m. *Daphne bholua* is mostly used in the paper making cottage industry for production of Desho "traditional paper". *D. bholua*

is the only *Daphne* species that is also used as an ornamental plant because of its strong fragrant flowers but people of longkhar harvest bark as raw materials for paper production only. The harvesting of daphne bark is done from April to October every year. It grows well in Longkhar having fertile soil and favorable climatic conditions.

The people of Longkhar collect daphne barks and supply them to the factories of Yangtse Gewog either in wet or dry form. In the past management cycle, they kept the *Daphne* plants/trees untouched to let the plants grow for high production in the coming management cycle.

2.5.4. Assessment of Resources

The rigorous assessment of *Daphne bholua* was carried out by adopting the methodology of DoFPs' Guidelines for Resource assessment and management of *Daphne*. This was done to review and validate the boundary prepared during participatory mapping, assess the availability of resources, and estimate the resources that can be harvested on sustainable basis from the demarcated bamboo blocks. While assessing, it was found that there were ample of resources in these blocks as illustrated briefly in the tables (3,4 &5).

Table 3. Number of plants with diameter of less than 10 cm at D10 (not harvestable)

Block (name)	Area (ha) (a)	No of sample plots (b)	Total no of plants of D10 < 10 cm (c)	Average no of plants of D10 < 10 cm per sample plot (=c/b) (d)	Average no of plants of D10 < 10 cm per ha (=100*d) (e)	Total no of plants of D10 < 10 cm in block (=a*e) (f)
I (Tsangnagpo)	202	15	189	12.6	1260.0	254520.0
II (Kingru)	197	15	201	13.4	1340.0	263980.0
III (Longkharwog)	28.3	9	68	7.6	755.6	21382.2
Total no of plants of D10 < 10 cm Diameter in all the blocks						539882.2

Table 4. Number of plants with diameter of > 10 cm at D10 (harvestable)

Block (name)	Area (ha) (a)	No of sample plots (b)	Total no of plants of D10 > 10 cm (c)	Average no of plants of D10 > 10 cm per sample plot (=c/b) (d)	Average no of plants of D10 > 10 cm per ha (=100*d) (e)	Total no of plants of D10 > 10 cm in block (=a*e) (f)
I (Tsangnagpo)	202	15	217	14.5	1446.7	292226.7
II (Kingru)	197	15	194	12.9	1293.3	254786.7
III (Longkharwog)	28.3	9	82	9.1	911.1	25784.4
Total no of plants of D10 > 10 cm Diameter in all the blocks						572797.8

Table 5. Available usable dry bark weight in blocks of daphne plants with diameter of > 10 cm

Block (name)	Area (ha) (a)	No of sample plots (b)	Total Wd (g) of plants with diameter of > 10 cm (c)	Average Wd per sample plot (=c/b) (d)	Average Wd(g) per ha (=100*d) (e)	Total Wd in block (=a*e) (f)	Total usable bark (53% of f)
SI (Tsangnagpo)	202	15	45570	3038	303800	61367600	32524828
II (Kingru)	197	15	37830	2522	252200	49683400	26332202
III (Longkharwog)	28.3	9	6642	738	73800	2088540	1106926
Total usable dry bark weight in all the blocks (g)							59963956
Total usable dry bark weight in all the blocks (kg)							59964
Annual Harvesting Limit (20% total usable dry bark weight) (Kg)							11992

2.5.5. Thoksumpa

Thoksumpa (*Paris polyphylla*) is a rhizomatous perennial plant, which can reach about one meter in height. It has spider-like flowers with long, thread-like, yellowish green petals, which are followed by small, scarlet berries in autumn. Satuwa grows best in broad-leaved forests bamboo forests, thickets, grassy or rocky slopes and stream sides at altitudes from 100 m up to 3500 m absl. Thoksumpa is considered as an important medicinal plant in China, India and Nepal as its rhizomes are said to have many healing properties and have shown antibacterial action against various bacteria. Furthermore, it is being used in treating fever and headache and healing burns and wounds.

A decade ago, there were plenty of Thoksumpa grown in the forest of Longkhar (Identified Thoksumpa blocks). But in past five to six years, in Bhutan, commercial interest in Thoksumpa grew with growing market demands fetching high prices in China, India and Nepal. This led to over exploitation of the resources with unsustainable harvesting by removing entire rhizomes which resulted in drastic decline of Thoksumpa resources in Longkhar forest. The group focused on reversing this trend for which they have been protecting the habitat without harvesting Thoksumpa in previous management cycle, consequently there are good regeneration.

2.5.6. Assessment of Resources

The rigorous assessment of *Thoksumpa* (*Paris polyphylla*) was carried out by adopting the methodology of DoFPs' Guidelines for Resource assessment and management of *P. polyphylla*. This was done to review and validate the boundary prepared during participatory mapping, assess the availability of resources, and estimate the resources that can be harvested on sustainable basis from the demarcated bamboo blocks. While assessing, it was found that there are limited harvestable *P. polyphylla* in the blocks, however, there are plenty of new regenerations which would become harvestable in next 6 to 7 years. Hence, the group was directed to nurture new regenerations and not to harvest for next 5 years.

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Table 6. Annual harvesting limit of Thoksumpa

Size of collection area in ha (or acres): a					312
Total sampling area covered in ha (total strip length (10.5=km in total) * strip width of 4m)/ 10,000: b					4.24
Total no of Thoksumpa plants	Number of Thoksumpa plants in sampling area (c)	Number of Thoksumpa plants per hectare (d) (= c * b)	Estimated total no of Thoksumpa in whole blocks (d*a)	AHL of Thoksumpa rootstocks in numbers (e) (= a *d)	AHL of Thoksumpa per Kg (= e * average weight of Thoksumpa rootstock (=120g)
Total no of plants flowering/ fruiting	2	8.48	2645.76	2645.76	317.4912
Total no of plants > 2 feet	5	30	9360	Non-harvestable Thoksumpa (young ones)	
Total no of plants 1 – 2 feet	27	162	50544		
Total no of plants < 1 feet	36	216	67392		
Total	70	420	127296		

2.5.7. Thrashing

In wood turning industries, the highly preferred raw materials for production of traditional wooden plates, cups mugs, etc. are wood burrs (special class (Dza) and normal (Baow)). Next to wood burrs, Thrashings are opted for its beautiful texture and pattern. With inferior quality when compared to wood burrs, the products of Thrashings are a bit cheaper than the products of wood burrs. The Shazops and wood burr collectors shared that, in near future wood burr resources are likely to exhaust, hence, thrashing would replace wood burr as raw materials for wood turning products. Thrashing bearing trees species include Persia species, Acer species and Betula species, which are found plenty in the forest of Longkhar.

2.5.8. Assessment of Resources

The abundance of Trashing species was assessed by counting and recording all the Trashing bearing species in Longkhar forest. Some sections of the forest are predominantly covered by Persia species, which are highly preferred species for wood turning. Next to Persia species, there are Acer and Betula species (which are also opted for wood turning) scattered all over Longkhar forest. Since there are plenty of Thrashing bearing species, the group has been recommended to fell 1 to 2 thrashing bearing species in a year, but with approval from Forest Range office, Dungzam, BWS.

2.6. Projection of Annual Harvest of NWFPs for commercial sales

As reflected in the table 7, the group projects to produce following quantities of NWFPs for selling products commercially in each season (annually). This quantity does not include those harvested for self-usage.

Table 7. Brief description of NWFPs for commercial harvesting in each season.

NWFP name	Harvesting area(s) (Blocks)	Parts harvested	Use purpose (traditional & others)	Harvesting season (months)	Estimated quantity for harvest in a season/year
Daphne	Block I	Barks	Traditional Paper	May-July	1700 kg
	Block III	Barks	Traditional Paper	May-July	1500 kg
	Block III	Barks	Traditional Paper	May-July	1100 kg
Bamboo (Matured/Tshar)	Block I	Culms (Pieces)	Bamboo products	February-March	15000 culms
	Block II	Culms (Pieces)	Bamboo products	February-March	17000 culms
Thoksumpa	Block I	Rhizome	Nil	Nil	Will not be harvested for another 5 years
	Block I	Rhizome	Nil	Nil	
Thrashing	Longkhar Forest	Base of the tree (Blocks)	For wood turning	October to March	600 pairs of wooden blocks

2.7. Harvesting prescriptions for sustainable NWFP harvesting

To make sure that resources are harvested scientifically and sustainably; following prescriptions will be strictly followed, which are based on the guidelines issued by DoFPS and/or based on local/scientific management of the species.

2.7.1. Daphne

- Cut the stem 10 cm above the ground to promote regeneration of shoots through a coppice system. As a rule of thumb this is at the height of a fist with the thumb up. The bark is peeled off after the stem is cut.
- Cut the stem slanting to avoid rotting of the remaining stump.
- Cut only stems of more than 3 cm in diameter at 10 cm from the ground; it is recommended to harvest as late as possible because the bark increment increases significantly over the years.
- Take care and do not to destroy/ damage remaining stands, seedlings, and saplings of Daphne while harvesting.
- Group should practice rotational harvesting method within collection area for sustainable resource management.

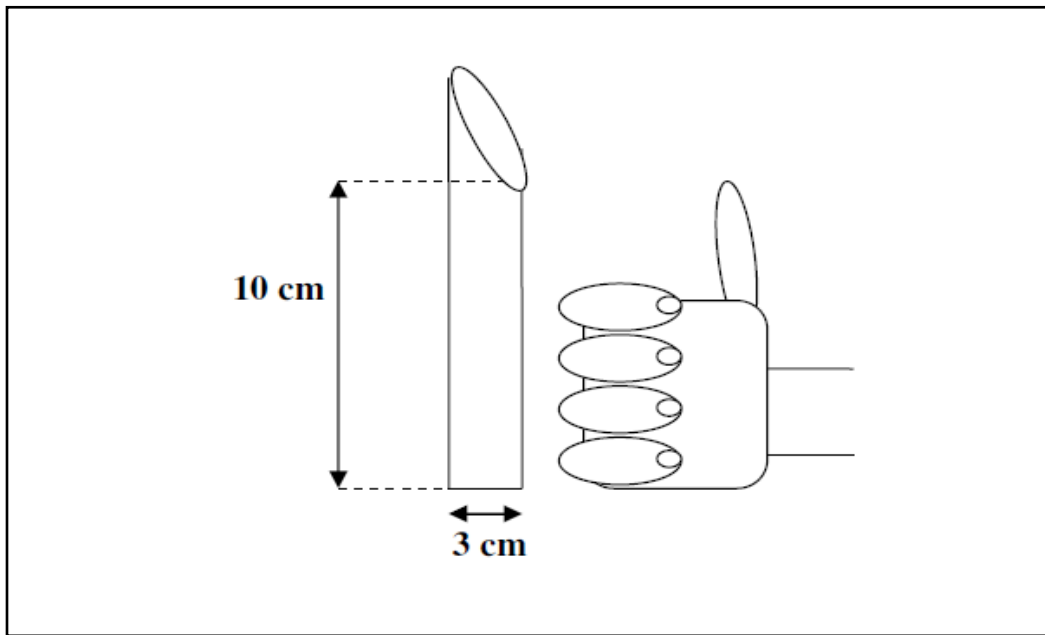


Figure 2: Minimal height from the ground and minimal diameter for cutting *Daphne*

2.7.2. Paris Polyphylla

- Harvest rootstocks of only mature *Thoksampa* that have provided seeds.
- Harvest rootstocks after seed dispersal (mostly in November but timing depends on the altitude/season).
- Plant a part of the rootstock with the roots and the upcoming sprout back into the soil- this part can grow into a new plant.



Figure 3. Place to cut *Thoksampa*: the left part with the roots and shoot is replanted, while the right part is collected.

2.7.3. Bamboo

- *Culms should be cut at the base or at the first long internodes above the node.*
- *Bamboo shoots or culms of less than one-year old should be left intact.*
- *Maximum 33 % of the 1+ year and 2+ year old culms per clump should be harvested each year and 67% should be kept for regeneration. This AHL can be applied to each available clump.*
- *70% of the 3+ years old culms can be harvested. It is important to leave some old culms to provide physical support to the 1+ year and 2+ years old culms.*
- *Culms should not be harvested when new shoots are sprouting as this practice will harm the new developing shoots. The preferred timing for harvesting bamboo culms is four to five months after the bamboo-shooting season and at least one to two months before the bamboo-shooting season (this season differs according to location and altitude).*
- *Harvesting of condensed/congested clumps should be done in a U-form or horseshoe form to maximize productivity. Young shoots only grow on the edge of the clump. Therefore, harvesting culms in a U-form will allow more shoots to grow.*

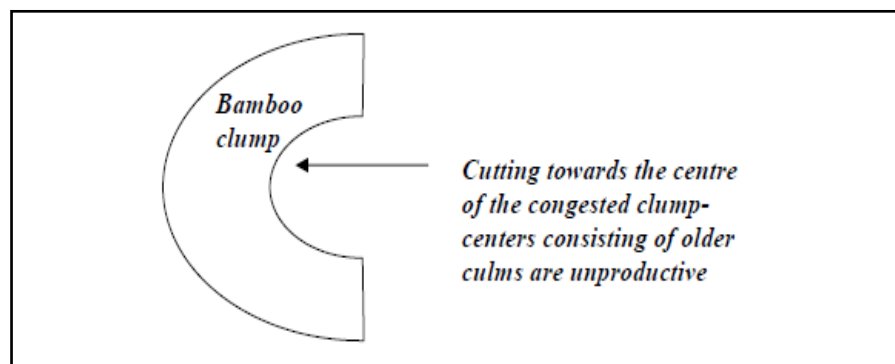


Figure 4. Bamboo cutting method.

2.7.4. Thrashing

Thrashings are beautiful-patterned wooden blocks, which are highly preferred next to wood burrs for production of various wood turning products. Though apparent in almost all the tree species, Thrashings are also rarely found like wood burrs. Thrashing collection and selling would be lucrative business for Longkhar people, however at the same time, it is important to harvest Thrashing sustainably. To achieve this, following prescriptions must be adopted while harvesting Thrashing.

- *Identify one to two thrashings in a year for harvest(felling).*
- *For sustainability of Thrashing, harvest only 1 to 2 numbers of Thrashing.*
- *Consult with the nearest forest office to check whether it is harvestable.*
- *Obtain permit from the forest office for felling and transport of products (Follow proper procedures)*

2.8. Marketing and sales plan

For the management period 2023 to 2027, the group will solely focus on harvest and huge production of Daphne barks as raw materials for making Desho Paper (Traditional Paper). This is to meet the high demand of paper factories within Yangtse and Bumdeling. The group will also focus on the diversification of bamboo products and marketing of these products to the Gewog, Dzongkhag, Neighbouring Dzongkhags and to AP state of India. In addition, this year, the group will also focus on the collection and supply of thrashing blocks to the wood turning industries. Refer table below.

Table 8. Marketing and sales plan of the group

NWFP	Potential market/buyer	Estimated quantity for sale per season (unit)	Form of Products (dry/fresh)	Expected price per unit	Total (Nu)
Daphne	Pvt. Paper Factories (from T/Y or other Dzongkhags) and Group Paper Factory (TY)	3600 kg	Dry	70-80/Kg	252000
Bamboo	Clums: Group members and other neighboring Villages	2Y+ =3600 1Y =10000	Freshly cut culms	0.5/Culms 1/Culms	1800 10000
	Products: Neighboring villages, TYangtse Dzongkhag and Other Dzongkhgs	Member sales their diversely made products and contribute Nu.3000 annually to the group fund.			36000
Thoksumpa	Other dzongkhags and neighboring province of China and States of India	For the next five years, the <i>Paris polyphylla</i> will not be harvest so to maximize new regenerations			Nil
Thrashing	Wood turning factories of Yangtse and Bumdeling Gewog	600 pairs of wood stumps/blocks	Wet and treated	Nu.500/pair	300000

2.9. Marketing arrangement

The potential markets for the sales of their raw materials and products was properly worked out. Therefore, the important aspect of this plan is to have well planned and strategic marketing arrangements. The harvesting of NWFPs and marketing of these products (both finished products and raw materials) will be organised and arranged by the groups led of Executive committee members. There will be equal sharing of profits after deduction of marketing expenses and contribution to the group fund.

Table 9. Procedures for NWFP harvesting, marketing, and sharing of income.

Activity	Procedures
Harvesting of products	All the group members shall participate during the harvest and collection of NWFPs.
Postharvest practices	The postharvest practices such as product diversification in terms of bamboo, and preparation of raw materials (Dry barks and wooden blocks) in terms of Daphne and Thrashing will be done involving all the members.
Marketing of products	The group shall appoint individuals who shall carry out marketing (One of the executive committee members (coordinator) with any member(s))
Income sharing	The profit from the sales of the products and raw materials will be shared equally among the group members. This will be divided after having contributed certain share to the group fund and clearing the various expenses made for the marketing
Raising of community development fund	Certain percent from the profit shall be deposited to the community development fund (Which shall be decided unanimously by the group members)

2.10. Benefits to the members

The profit-sharing mechanism is such that, the 20% of revenue generated through sales of products (after deducting various expenses) would be deposited to the group fund while remaining 80% of the revenue will be divided amongst the group members. This will help in boosting the revenue of the group as well as help prospering the individual member. Table 10 shows the financial projection of the benefits to the group members.

Table 10. Financial projection and benefits to the members from NWFP sales

Sl. No	Particulars	Products (Names)				Total amount (Nu.)
		Bamboo	Daphne	Thoksumpa	Thrashing	
1. Revenues						
1.a	Total quantity (unit)	14500 (culms)	3600 (kg)	Nil	600 (pairs)	-
1.b	Market price (Nu per unit)	1	70		500	-
1.c	Total revenues from forest product sales (a*b)	14500	252000	0	300000	
A	Total revenues					566500
2. Expenses						
2.a	Packaging materials	1000	1000	0	2000	4000
2.b	Royalty (20% of total sales/Revenue)- Contribution to Community Development Fund	2900	50400		60000	113300
2.c	Transportation costs	6000	6000		6000	18000
2.d	Compensation fee to executive committee/members for marketing products (2% of total Sales/Revenue to be divided among marketing group)	435	7560	0	9000	16995
2.e	Incidental expenses	2500	2500		2500	7500
B	Total expenses					159795
C	Total benefit (net-income) of group members (Total revenues A – Total expenses B)					406705
D	Average income to group members (C/ no of group members) in one year					36973
E	Average income to group members (C/ no of group members) in 5 years					184866

2.11. Benefits to the group

The natural resources management and marketing mainly focuses on generating income for the group as well as for the individual members, but keeping in mind the sustainable techniques are being followed while harvesting of resources-this ensures the sustainability of the resources for next 10 to 20 years. The group strategized the profit-sharing mechanism, whereby the 20% of revenue generated through sales of products (after deducting various expenses) would be deposited to the group fund while remaining 80% will be divided among the group members. Table 11 shows the financial projection of the benefits to the group.

Table 11: Financial projection of the benefits to the group

Sl. No	Particulars	Amount (Nu)
Revenues		
1	Membership fees	0
2	Contribution to the community development fund (CDF) from sales of Bamboo products (20 % of the sales)	2900
3	Contribution to the CDF from sales of Daphne raw materials (20 % of the sales)	50400
4	Contribution to the CDF from sales of Thrashing (20 % of the sales)	60000
A	Total revenues	113300
Expenses (only include expenses that will be paid from the group funds)		
1	community infrastructure development	20000
2	working expenses	10000
B	Total expenses	30000
C	Total Benefit to Group (A-B) in a year	83300
D	Total Benefit to Group (A-B) in 5 years	416500

2.12. Annual Work Plan

ACTIVITY		TIMEFRAME				
		Year I	Year II	Year III	Year IV	Year V
Collect NWFP resources as per harvesting guidelines	a. Daphne (bark)	May-July	May-July	May-July	May-July	May-July
	b. Thrashing (wooden blocks)	Oct-March	Oct-March	Oct-March	Oct-March	Oct-March
	c. Bamboo (culms)	Feb-April	Feb-April	Feb-April	Feb-April	Feb-April
	d. Thoksumpa (rhizome)	will be left untouched for next 4 to 5 years for better production in future				
Carry out waste cleaning in the forest as well in the community		4 times a year	2 times a year	4 times a year	2 times a year	4 times a year
Raise saplings in the nursery through seeds and vegetation propagation	a. Thoksumpa through rhizomes	300 saplings	300 saplings			
	b. Daphne through seeds	500 saplings	500 saplings			
	c. Wood turning species (<i>Persia spp.</i> , <i>Acer spp.</i> , <i>Betula spp.</i> etc.) through seeds	300 saplings each	300 saplings each			
	d. Timber species like <i>Pine</i> , <i>Cypress</i> , <i>Champ</i> etc.) through seeds	500 saplings each	500 saplings each			
Carry out plantation				April-June	April-June	April-June
Conduct meetings	a. General meeting	Jan and July	Jan and July	Jan and July	Jan and July	Jan and July
	b. Adhoc meeting/committee meeting	As and when required	As and when required	As and when required	As and when required	As and when required
Carry out monitoring	a. by group	March and Sept	March and Sept	March and Sept	March and Sept	March and Sept
	b. Jointly by group and BWS	Dec	Dec	Dec	Dec	Dec

2.13. Monitoring and Evaluation

2.13.1. Monitoring by Group lead by Committee members

Monitoring will be conducted at least 2 times a year but if there are genuine requirements to increase the frequency of the monitoring, it shall be done based on the decision of the committee members. The main motive of the monitoring will be to patrol whole Longkhar forests there by collecting information regarding resource management status, threats (biotic and abiotic) to NWFPs and Forests, illegal activities, etc. Right after the monitoring, the team shall present the report to the Group members and come up with way forwards and various remedies to the challenges if encountered any. Regarding the encounter of any illegal activity, the group shall report to Dungzam Range office and sort out.

2.13.2. Joint Monitoring by Group and BWS

The joint monitoring will be carried out once a year, during the time when BWS office comes for monitoring and evaluation of Longkhar NWFP Management and Marketing Group. Right after evaluating the administrative, financial works in the group office, the group members shall accompany forestry officials to various NWFPs blocks for monitoring of NWFPs harvesting sites. In the joint monitoring, the forestry officials will assess whether the group has carried out activities as per the management plan prescription.

PART 3: By-Laws of NWFP Management and Marketing Group

3.1. Membership arrangement and resignation

- All individuals/households of Longkhar Community, with traditional rights for obtaining forest produces from Longkhar forests, are the member of Longkhar NWFP Management and Marketing Group. There are, so far, 11 members in the group. Any individual/household apart from Longkhar community, will be disallowed to join the group.
- In the case of a member wishing to withdraw from the group, he or she shall do so but only upon the approval from the management committee members, who will decide upon discussion with all the members.
- If the reasons aren't genuine and unavoidable, the member opting to resign will be given condition that he/she shall resign with no benefits/claim from the group fund and shall be forfeited from the rights of collection of any non-wood forest resource. In other words, the member will be treated similar as outsider and will be levied permit fees and royalty as equal as outsiders.
- Upon thorough investigation, if the reasons are found genuinely unfortunate, the committee members shall approve the resignation by giving his or her share of benefits from the group fund till the date he/she puts up the resignation.

3.2. Members (ToR)

The members shall:

- Be responsible for sustainable management of the NWFP Management Unit, in accordance with the approved management plan and by-laws.
- Ensure that no person collects, use, remove, damage, or destroy any forest produce from the NWFP Management Unit without prior authorization.
- Actively participate in NWFP management and marketing planning and decision-making (should be present in all the meetings).
- Take part in the election of Executive Committee members (Elect good leaders).
- Cooperate with the Executive Committee Members in carrying out forest management and marketing operations (especially by contributing manpower if required).
- Collaborate with Executive Committee Members in regulating FNCRR, Management and Marketing plan rules and regulations and By-law.
- Involve in all the activities as per the annual work plan.

3.3. Executive committees (ToR)

3.3.1. Chairperson

Name and contact No: **Ugyen Tenzin (77806101)**

The chairperson shall:

- Look after overall administrative works.
- Chair, coordinate and facilitate all meetings related to NWFP management and Marketing.
- Have the authority to accord approval for forest products collection and accordingly issue permit (with Secretary).
- Arrange and coordinate marketing of NWFPs (by maintaining good contacts with the buyers, traders, and exporters).
- Countersign financial transaction and authorization (with Treasurer and Secretary).
- Organize field visits annually for monitoring and evaluation.
- Resolve disputes and conflicts among members.

3.3.2. Secretary

Name and Contact No: **Shacha Tshering (17567490)**

The secretary shall:

- Assist and render necessary support to the Chairperson.
- Officiate Chairperson during the absence of the Chairperson.
- Manage the correspondence (writing of letters and keeping minutes of meetings).
- Maintain the files and keep good records of the matters pertaining to the group.
- Approve and issue forest products (with Chairperson).
- Prepare Annual Progress Report (with other NWFP Management Committee members).
- Arrange and provide good hospitality for the visitors.
- Counter-sign financial transaction and authorization (with Treasurer and Chairperson).
- Arrange and provide good hospitality for the visitors.

3.3.3. Treasurer

Name and contact No: **Gonpo Wangdi (17561294)**

The Treasurer shall:

- Administer financial arrangements.
- Keep records of financial transactions, account, and expenditure.
- Collect membership fees, permit fees and other fees.
- Prepare and present financial report/status to the members during annual meeting.
- Countersign financial transaction and authorization (with Chairperson and Secretary).
- Assist Chairperson and Secretary in marketing.

3.4. Tenure of committee member

The committee members elected from the group members, by the group members unanimously, will serve for 5 years which is as same as the term for this management plan (2023-2027)

3.5. Incentives to the committee members

Since the group is at its infancy stage, there are still needs to accumulate as much funds as possible. Therefore, apart from the necessary expenses which is to be made during the execution of any group related work, the executive committee members will not be provided with incentives.

3.6. Sources for generation of group fund

Sources of generation of group fund will be from membership registration fees, annual contribution & collection from sale proceeds as detailed below.

- The membership registration fee was already collected during the formation of this group. Therefore, since all the households of Longkhar Community are the members of this group and no outsiders will be included, the collection of registration fee is out of discussion.
- However, Nu.3000 will be collected from all the members annually to meet different expenses for implementing group related activities and to raise group funds.
- In addition to this collection, the group generate income mainly through the issuance of non-wood forest resources to both members and outsiders by levying permit fees and royalties, 100% of this income will go into the group account as sale proceed.

3.7. Group fund management and utilization

The group already has the group account; therefore, all the funds/budgets of the group are kept in the group account. The executive members will have the authority to deposit or withdraw funds/budget. The ceiling of amount to be withdrawn from the group account for office and administrative expenses, must be decided by the executive committee through discussion with the group members. The percentage of fund to be used for different purpose are as follows:

- NWFP resource development: For sustainability of NWFPs, various resource development activities need to be carried out. Therefore, annually, certain percent of amounts to be allocated for carrying out these activities. The amounts to be decided unanimously as per the actual expenses incurred.
- Loan: To improve the economic condition of the group members, the loan will be sanctioned to the members by assessing and confirming genuine needs of the applicants, however, while giving loan, the loan out amount should not exceed 50% of the total fund of the group. The terms and conditions and interest of loan should be decided by all the members.
- Welfare to members during natural calamities: In the event of destruction of properties or death by natural calamities like earthquake, floods, windstorm, house fire, etc. certain welfares need to be taken care of. During this misfortune, by thoroughly accessing the severity of the cases (to be done all the members lead by committee members), the affected

households will be provided Nu. 30,000/ if the case is severe, Nu. 20000/- if the case is moderate and Nu.10,000 to the less severe cases.

3.8. Conflict resolution

- In case disputes or conflicts among communities arise about access to certain areas for NWFP harvesting; the matter shall be put to the Gewog Tshogchung (GT) Dzongkhag Tshogdue (DT) for further decision.
- Conflicts or disputes within the NWFP Management Group, which cannot be resolved by the joint executive committee members shall be forwarded to the Gewog Administration and Bumdeling Wildlife Sanctuary.
- In the event there is a conflict between the NWFP Group and outsiders, the case shall be forwarded to the and Gewog Administration and Bumdeling Wildlife Sanctuary.

3.9. Meeting and purpose

3.9.1. Committee meeting:

The committee meetings shall be called as and when required.

3.9.2. General meeting:

The general meetings will be conducted for 2 times in a half yearly basis. This is generally to discuss regarding the activities and achievements of the plan and to resolve various issues.

3.10. Penalties

Fines and penalties will be imposed for the following scenario:

- It is mandatory for all the members to pay annual fee of Nu. 3000. If the member fails to pay on stipulated time, the members are liable to pay penalty of Nu.1500 in addition to annual fee. (ie. Nu.3000+Nu.1500)
- All members should turn up for any sort of meetings and works. If the member is absent for the meeting/work he is liable to pay Nu.1000 and if the member fails to show up on time, he/she will be fined Nu.200/-
- In case a committee member misuses the ground fund, he or she will be liable to pay double the misused amount and will be expelled from the role of committee member.
- If a member is found not obeying to committee members for the work mentioned in the plan, he or she shall be liable to pay Nu.1,000/-
- In case a member of the NWFP group violates the harvesting rules as set in this plan, he or she will not be allowed to harvest the NWFP anymore for the rest of the season and the next season.

3.11. Permit Fees and Royalty

Products	Quantity Unit	Group Members	Outsider
		Rate (Nu)	Rate (Nu)
Bamboo	1 Pcs	0.5	As per market rate
Tsar	1 Pcs	1	
Thoksampa	1 Kg	750=wet 1500=dry	As per prevailing rate
Daphne	1 Kg	15	As per market rate
Thrashing	1 piece	25	Not allowed for outsiders

3.12. Schedule of Fines and Penalty

Types of Offence	Fines for Members and Outsider	Compensation
Collection of matured bamboos from Management Area without collection permit	Nu. 50 per piece member, Nu.50 per piece	Compensation at fair market value decided by NWFMG
Collection of immature bamboos (Tsar) from Management Area without collection permit	Nu. 60 per piece, Nu.60 piece for outsider	Compensation at fair market value decided by NWFMG
Collection of Daphne bark from Management Area without collection permit	Nu. 100/Kg in dry for member, 150/kg for outsider	Compensation at fair market value decided by NWFMG
Collection of Thrashing from Management Area without collection permit	Nu. Nu. 500/Pairs for member, Nu.800/Pairs for outsider	Compensation at fair market value decided by NWFMG

3.13. Powers of the DoFPS

The department has the right to terminate the NWFP management and marketing plan in case of any violation of the management plan prescription or in the event of inability to carry out the activities as per the plan.

Acronyms

AWP	Annual Work Plan
CFO	Chief Forestry Officer
BWS	Bumdeling Wildlife Sanctuary
DoFPS	Department of Forests and Park Services
FNCA	Forest and Nature Conservation Act
FNCRR	Forest and Nature Conservation Rules and Regulations, 2017
GPS	Global Positioning System
GIS	Geographic Information System
Ha	Hectare
Masl	Meter above sea level
MoENR	Ministry of Energy and Natural Resources
NWFPMG	Non-Wood Forest product Management Group
NWFMP	Non-Wood Forest Management Plan
NWFP	Non-Wood Forest Produce
RGoB	Royal Government of Bhutan
SFED	Social Forestry and Extension Division
SRF	State Reserved Forest
AHL	Annual Harvesting Limit
G	Gram
Kg	Kilogram

Appendix I: Resources assessment data collection sheets

Form I: Block description form for Bamboo (*Borinda grossa*)

Block name:

Access	<i>What is the walking distance from the nearest village to the block? (Shade)</i>		
	Less than 3 hours	Within a day	More than a day
	<i>What is the walking distance from the block to the nearest road? (Shade)</i>		
	Less than 3 hours	Within a day	More than a day

Slope	Very steep	Steep	Moderate	Gentle
	>100%	70-100%	45-70%	<45%

Forest description	Canopy cover:
	Dominant Species:
	Associate species:

Sub-block	No.	Area (Ha)	Productive area (Ha)	No. of sample plots (5m x 5m)
	1			
	2			
	3			

Bamboo Flowering	Sporadic	Gregarious	Last time of mass flowering:	None

<i>Plantation Potential</i>	Sub-blocks	Has plantation potential	No potential	Not applicable
	1			
	2			
	3			

<i>Current management</i>	Current Management practices in entire block:
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<i>Current threats</i>	Threat	Description
	Fire	
	Grazing	
	Illegal exploitation	
	Over-exploitation	
	Other;	

<i>Opportunities</i>	Future opportunities for the entire block:
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Form II: Bamboo resource assessment in the field

Date:..... Village:..... Recorder

Name(s):.....

Block:..... Sub block:..... Local Name of RA

Area..... Slope(%).....

Plot No:	Altitude	Total Clump	Clump condition *	Culms harvested over last 5years	Bamboo stock			Shoots (<1year culms)	Harvestable culms**			Total no of culms
					1+ year (No)	2+ years (No)	3+ years (No)		1+ year (No)	2+ years (No)	3+ years (No)	

* Clump condition: Good, damaged, congested, or flowering

** Harvestable culms have a diameter of 2-4cm, a length of about 5m, are healthy and straight and are fresh green and relatively soft (Counting of harvestable culms is for checking and balancing purpose only and should not be used in the AHL formula.

Form I: Block description form for Thoksumpa (*Paris polyphylla*)

Date:

Collection site name:

Access	<i>What is the walking distance from the nearest village to the block?</i>		
	Less than 3 hours	Within a day	More than a day
	<i>What is the walking distance from the block to the nearest road?</i>		
	Less than 3 hours	Within a day	More than a day

Slope	Very steep	Steep	Moderate	Gentle
	>100%	70-100%	45-70%	<45%

Forest description	Canopy cover:
	Dominant Species:
	Associate species:

Productivity	Estimated quantity that can be collected from the collection site:
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<i>Current management</i>	Current Management practices in entire block:
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<i>Current threats</i>	Threat	Description
	Fire	
	Grazing	
	Illegal exploitation	
	Over-exploitation	
	Other;	

<i>Opportunities</i>	Future opportunities for the entire block:
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Form II: Thoksumpa (*Paris polyphylla*) resource assessment form

Date:.....Village:.....Recorder Name(s):.....

Block:.....Sub block:.....Local Name of RA Area:.....

Slope(%).....Coordinates of Transact Start point: End Point.....

[illegible]

Form I: Block description form for Daphne (*Daphne bholua*)

Block name/No:

Access	<i>What is the walking distance from the nearest village to the block? (Shade)</i>		
	Less than 3 hours	Within a day	More than a day
	<i>What is the walking distance from the block to the nearest road? (Shade)</i>		
	Less than 3 hours	Within a day	More than a day

Slope	Very steep	Steep	Moderate	Gentle
	>100%	70-100%	45-70%	<45%

Forest description	Canopy cover:
	Dominant Species:
	Associate species:

Sub-block	No.	Area (Ha)	Productive area (Ha)	No. of sample plots
	1			
	2			
	3			

	Sub-blocks	Has plantation potential	No potential	Not applicable
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Plantation Potential	1			
	2			
	3			

Current management	Current Management practices in entire block:
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Current threats	Threat	Description
	Fire	
	Grazing	
	Illegal exploitation	
	Over-exploitation	
	Other;	

Opportunities	Future opportunities for the entire block:
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Form II: Data Collection sheet for Daphne

Date:

Recorder name:

Block:

Sub-block:

Local name of the area:

Slope:

Sample Plot:

Plant Serial No	<i>No of stems at D10 (Nos)</i>	<i>D10 per stem (cm)</i>	<i>Conversion to single stem*</i>	<i>Remarks</i>

* D10 (Día at 10 cm from the ground)

* Conversion of multi-stemmed plants into a single stem according to the quadratic mean formula, see box 3 in paragraph 2.2; the conversion can be done at the time of analysis.

Appendix II: Shapefiles

(Find the attached shape file)